

GADAG, India

WMO: 432010

Lat: 15.4175N

Lon: 75.6444E

Elev: 650

StdP: 93.76

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.6	16.6	-0.5	3.9	28.7	1.9	4.7	28.7	7.2	25.5	6.3	25.8	1.0	140	0.284

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.8	37.6	20.4	36.7	20.4	35.8	20.5	25.4	30.2	24.7	29.3	24.2	28.7	1.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.7	27.5	23.6	19.9	26.9	23.0	19.2	26.4	82.2	30.8	79.3	29.2	76.9	28.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.2	5.4	DB	13.0	38.8	1.0	0.9	12.2	39.5	11.6	40.0	11.1	40.5	10.3	41.2	(4)
(5)			WB	8.5	26.6	0.9	1.1	7.9	27.4	7.3	28.1	6.8	28.7	6.2	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	23.6	26.0	28.7	29.9	29.3	26.2	25.0	24.8	25.1	25.5	24.3	23.0	(6)
(7)		DBStd	2.52	1.37	1.59	1.54	1.28	1.48	1.55	1.09	1.05	1.15	1.19	1.31	1.42	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5814	421	448	578	596	599	486	463	457	454	479	429	404	(10)
(11)		CDD18.3	2772	162	214	320	346	341	236	205	199	204	221	179	146	(11)
(12)		CDH23.3	24912	1367	2252	3957	4343	3919	1688	1054	997	1231	1656	1337	1111	(12)
(13)		CDH26.7	10809	438	1065	2181	2478	2118	566	220	202	332	547	381	281	(13)
(14)	Wind	WSAvg	2.2	1.6	1.6	1.6	1.9	2.5	3.5	3.6	2.9	2.2	1.6	1.8	1.7	(14)
(15)	Precipitation	PrecAvg	669	1	3	14	41	81	86	82	87	110	113	43	8	(15)
(16)		PrecMax	1052	11	20	217	119	142	254	266	330	314	399	188	90	(16)
(17)		PrecMin	393	0	1	0	12	9	20	23	24	21	3	2	0	(17)
(18)		PrecStd	163	2	4	41	28	33	54	51	67	69	81	59	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	35.3	37.5	38.9	38.7	35.1	31.2	31.2	32.4	32.4	31.4	31.5	(19)
(20)			MCWB	18.2	18.6	19.8	20.5	19.9	22.2	22.0	21.9	20.7	19.8	19.5	19.2	(20)
(21)		2%	DB	31.0	34.2	36.5	37.9	37.6	32.9	29.9	29.8	31.0	31.4	30.5	30.1	(21)
(22)			MCWB	17.7	18.4	19.9	20.5	20.5	22.8	22.0	22.1	21.1	19.8	19.5	18.7	(22)
(23)		5%	DB	30.1	33.1	35.7	37.0	36.5	31.3	29.0	28.8	29.8	30.6	29.7	29.2	(23)
(24)			MCWB	17.4	18.3	19.7	20.4	20.9	22.9	22.2	22.1	21.6	20.0	19.5	18.0	(24)
(25)		10%	DB	29.2	32.0	34.8	36.0	35.2	29.8	27.9	27.8	28.7	29.7	28.9	28.3	(25)
(26)			MCWB	17.4	18.1	19.2	20.6	21.3	22.9	22.2	22.1	21.8	20.4	19.3	17.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	23.7	26.2	26.4	26.3	25.4	24.5	24.6	24.4	24.5	23.7	23.0	(27)
(28)			MCDWB	27.9	30.3	32.9	32.8	30.8	29.1	27.2	27.4	27.4	28.2	27.5	27.7	(28)
(29)		2%	WB	21.1	22.3	24.5	25.1	25.3	24.6	23.6	23.7	23.7	23.5	22.7	21.8	(29)
(30)			MCDWB	26.7	29.4	30.5	30.5	30.1	28.6	26.8	26.9	27.2	27.2	26.6	26.5	(30)
(31)		5%	WB	20.1	21.1	23.4	24.2	24.7	24.0	23.1	23.1	23.2	23.0	22.0	20.9	(31)
(32)			MCDWB	26.6	27.8	29.2	29.4	29.5	28.1	26.4	26.6	26.8	26.5	26.1	25.4	(32)
(33)		10%	WB	19.2	20.2	22.4	23.6	24.0	23.6	22.7	22.6	22.7	22.5	21.5	20.1	(33)
(34)			MCDWB	26.0	27.5	28.5	28.9	28.9	27.6	25.9	26.0	26.2	25.9	25.6	25.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	12.3	12.4	12.8	12.1	7.4	6.1	6.3	7.3	8.4	9.7	11.2	(35)
(36)			MCDBR	12.6	13.1	13.0	13.7	13.7	9.8	7.8	8.0	9.3	10.0	10.7	12.1	(36)
(37)			MCWBR	5.0	5.2	5.2	4.7	4.1	2.7	2.2	2.4	2.6	3.6	4.2	4.8	(37)
(38)		5% WB	MCDWR	10.6	11.8	12.2	12.6	11.7	8.0	6.1	6.3	6.9	7.5	8.0	9.5	(38)
(39)			MCWBR	5.2	5.3	5.5	4.9	4.3	3.3	2.7	3.1	3.0	3.3	3.7	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.412	0.431	0.505	0.566	0.554	0.505	0.510	0.490	0.469	0.513	0.460	0.430	(40)	
(41)		taud	2.208	2.177	1.986	1.873	1.926	2.114	2.095	2.164	2.230	2.048	2.153	2.185	(41)	
(42)		Ebn at Noon	875	876	820	768	766	796	794	818	839	790	821	844	(42)	
(43)		Edh at Noon	138	149	185	208	194	159	163	153	143	168	145	139	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.27	6.06	6.53	6.72	6.52	4.95	4.28	4.40	4.78	4.99	4.91	4.97	(44)	
(45)		RadStd	0.22	0.21	0.25	0.24	0.30	0.38	0.33	0.24	0.40	0.26	0.32	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.14	N/A	+0.61	+0.34	+0.39	+0.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page